

Mudit Sharma

PhD Student in Computer Science and Engineering

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Research Profile

PhD researcher in Computer Science and Engineering at the **Indian Institute of Technology Ropar (IIT Ropar)**, supervised by **Dr. Shweta Jain**. My research focuses on optimization and sequential decision-making, particularly multi-armed bandits, dueling bandits, and nonconvex optimization. I aim to develop theoretically principled and computationally efficient algorithms for learning and decision-making under uncertainty.

Research Interests

Online Learning · Multi-Armed Bandits · Dueling Bandits · Nonconvex Optimization

Education

Ph.D. in Computer Science and Engineering

Indian Institute of Technology Ropar, Punjab, India

2025–Present

CGPA: 8.47/10

• Advisor: Dr. Shweta Jain

Bachelor of Technology (B.Tech.) in Computer Science and Engineering

Chandigarh University, Punjab, India

2020–2024

CGPA: 7.81/10

Research Experience

Junior Research Fellow

Supervisor: Dr. Shweta Jain

IIT Ropar

August 2024–June 2025

- Developed and analyzed algorithmic approaches for dueling multi-armed bandits using preference-based, pairwise-comparison feedback.
- Investigated exploration–exploitation strategies and regret-minimization methods for sequential decision-making under uncertainty.
- Formulated research questions and evaluated candidate algorithms through theoretical analysis and reproducible computational experiments.

Publications

Peer-Reviewed Publication

1. **Mudit Sharma**, Shweta Jain, Vaneet Aggarwal, and Ganesh Ghalme. “Lipschitz Dueling Bandits over Continuous Action Spaces.” Accepted at *Advances in Neural Information Processing Systems (NeurIPS 2026)*; to appear. **CORE Rank: A***.

Teaching and Mentoring

Teaching Assistant, GE 103: Introduction to Programming

Instructors: Dr. Shweta Jain and Dr. Shashi Shekhar Jha

IIT Ropar

Summer 2025

- Mentored students in core C programming concepts and guided them through problem-solving and debugging exercises.

Teaching Assistant, AI111: Mathematical Foundations of AI

Instructors: Dr. Puneet Kumar and Dr. Shweta Jain

IIT Ropar

Winter 2025

- Guided students through mathematical concepts underpinning artificial intelligence and supported their application to foundational problems.

Teaching Assistant, CS203: Data Structures

Instructors: Dr. Shweta Jain and Dr. Swapnil Dhalmal

IIT Ropar

Summer 2026

- Facilitated learning of data structures and algorithms through hands-on guidance in implementation, debugging, and complexity-aware problem solving.

Technical Skills

Programming Python, C++, Java

Research expertise Multi-armed bandits, dueling bandits, stochastic and nonconvex optimization, sequential decision-making, exploration–exploitation, regret analysis, and algorithm design and analysis

Tools Linux, LaTeX, Docker, AWS

Languages English (professional), Hindi (native), Punjabi (professional)

Achievements

Qualified Graduate Aptitude Test in Engineering (GATE)

2024, 2025